Fee Concentration Insurance

Mathematical Specification

ThetaSwap

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Abstract

We specify an insurance mechanism against adverse fee concentration in Uniswap V4 liquidity provision. The fee concentration index $A_T = (\sum_k \theta_k \cdot x_k^2)^{1/2}$ measures realized fee inequality across LP positions. The Ma–Crapis competitive null $A_T^{1/N} = \sqrt{\Theta/N^2}$ provides the equal-share baseline. The concentration deviation $\Delta^+ = \max(0,\, A_T - A_T^{1/N})$ captures excess concentration above the competitive equilibrium and maps to price space via the odds ratio $p = \Delta^+/(1-\Delta^+)$.

Econometric identification (quadratic deviation exit hazard model) establishes an inverted-U relationship: for the pool we analyzed (ETH/USDC 30bps), below the turning point $\Delta^* \approx 0.09$, concentration is protective (shelter effect); above it, concentration drives LP exits[1]. We find contingent empirical demand from passive liquidity providers for insurance against adverse competition, and that the optimal payoff function is quadratic. The turning point Δ^* calibrates insurance demand.

The hook tracks three co-primary state variables—A_T (fee concentration index), Θ (aggregate turnover rate $\sum_k \theta_k$), and N (active position count)—with Δ^+ derived on the fly. The mechanism is perpetual with jump-adjusted funding and USDC numeraire.

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1 Fee Concentration Index

1.1 Primary Index: A_T

The fee concentration index A_T is a path-dependent, HHI-weighted measure of adverse competition in liquidity provision. It is the *primary* computed and tracked quantity:

$$A_T = \Big(\sum_k \theta_k \cdot (x_k)^2\Big)^{1/2} \tag{1}$$

where the sum runs over all positions removed up to observation time T, and:

- $x_k = \dfrac{\ell_k}{L(p^*)}$ is the fee share captured by position k, where ℓ_k is the position's liquidity and $L(p^*)$ is the total liquidity in the active tick range;
- $\theta_k = \dfrac{1}{\Delta B_k}$ is the sophistication weight, where $\Delta B_k = \text{block(removal)} - \text{block(creation)}$ is the position lifetime in blocks;
- $\theta_k = 1$ for JIT positions ($\Delta B_k = 1$ block: same-block add and remove).

The index satisfies $A_T \in [0, 1]$.

1.2 Co-Primary State Variables

The hook tracks three state variables at each liquidity event:

Definition 1.1 (Co-Primary State)**.**

$$A_T = \Big(\sum_k \theta_k \cdot x_k^2\Big)^{1/2} \qquad \textit{fee concentration index} \tag{2}$$

$$\Theta = \sum_k \theta_k \qquad \textit{theta sum (aggregate turnover rate)} \tag{3}$$

$$N = |\{k : \textit{position}_k \ \textit{active}\}| \qquad \textit{active position count} \tag{4}$$

All three are stored on-chain. The derived quantities $A_T^{1/N}$ and Δ^+ are computed on the fly from these three values.

The Ma–Crapis [MC24] symmetric equilibrium assumes all positions capture equal fee shares $x_k = 1/N$. Under this null hypothesis:

$$A_T^{1/N} = \sqrt{\frac{\Theta}{N^2}} \tag{5}$$

Proposition 1.2 (Null as Lower Bound)**.** *For any fee share distribution $\{x_k\}$ with $\sum_k x_k = 1$ and $x_k \geq 0$:*

$$A_T \geq A_T^{1/N} \tag{6}$$

with equality if and only if $x_k = 1/N$ for all k.

Proof. Under equal shares $x_k = 1/N$:

$$(A_T^{1/N})^2 = \sum_k \theta_k \cdot \frac{1}{N^2} = \frac{\Theta}{N^2}$$

For arbitrary shares, by Jensen's inequality on the convex function $x \mapsto x^2$:

$$\sum_k \theta_k x_k^2 \geq \frac{\Theta}{N^2} \quad \text{when} \ \sum_k x_k = 1$$

since $\sum_k x_k^2 \geq 1/N$ by the constraint $\sum x_k = 1$ (this is the HHI lower bound). Multiplying each term by θ_k/Θ and using convexity preserves the inequality. Taking square roots gives $A_T \geq A_T^{1/N}$. □

1.3 Concentration Deviation: Δ^+

The economically meaningful treatment variable is the excess concentration above the competitive null:

Definition 1.3 (Concentration Deviation)**.**

$$\Delta^+ = \max\big(0,\ A_T - A_T^{1/N}\big) \tag{7}$$

By theorem 1.2, $\Delta^+ \geq 0$ always. $\Delta^+ = 0$ corresponds to the competitive equilibrium (all LPs earn equal fee shares). $\Delta^+ > 0$ indicates excess concentration: a few LPs capture disproportionate fee revenue.

1.4 Price Mapping

Definition 1.4 (Concentration Price)**.** *The price of fee concentration in numeraire (USDC) is the odds ratio:*

$$p = \frac{\Delta^+}{1 - \Delta^+} \tag{8}$$

with $p \in [0, \infty)$ for $\Delta^+ \in [0, 1)$. The inverse mapping is $\Delta^+ = p/(1 + p)$.

Proposition 1.5 (Monotonicity)**.** *$p(\Delta^+)$ is strictly increasing on $[0, 1)$.*

Proof. $\dfrac{dp}{d\Delta^+} = \dfrac{1}{(1 - \Delta^+)^2} > 0$ for all $\Delta^+ < 1$. □

2 Econometric Calibration

The econometric identification (see companion document `lp-insurance-demand.tex`) establishes the demand structure for fee concentration insurance.

2.1 Quadratic Deviation Model

The exit hazard model with quadratic treatment:

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \cdot \text{dev}_t + \beta_2 \cdot \text{dev}_t^2 + \beta_3 \cdot \text{IL}_t + \beta_4 \cdot \ln(\text{age}_{i,t})\big) \tag{9}$$

where $\text{dev}_t = A_T - A_T^{1/N}$ is the deviation from the competitive null.

2.2 Key Results

Lag	β_1 (linear)	p	β_2 (quadratic)	p
1	-23.18	0.012^{**}	$+129.20$	0.030^{**}
2	-43.42	0.016^{**}	$+226.92$	0.065^{*}
3	-32.44	0.001^{***}	$+205.34$	0.004^{***}

2.3 Inverted-U and the Insurance Trigger

The model produces an inverted-U with turning point:

$$\Delta^* = -\frac{\beta_1}{2\beta_2} \approx 0.09 \tag{10}$$

- **Below Δ^* (shelter regime):** Mild fee concentration correlates with high volume. All LPs benefit from elevated fee revenue. $\partial P(\text{exit})/\partial \text{dev} < 0$.
- **Above Δ^* (Capponi regime):** Extreme concentration means passive LPs' effective fee rate drops below their exit threshold. $\partial P(\text{exit})/\partial \text{dev} > 0$. Capponi & Zhu (2024): $\partial p_U^*/\partial \varphi > 0$, so lower $\varphi \to$ exit.

The turning point $\Delta^* \approx 0.09$ identifies when insurance demand activates. This calibrates the protection mechanism regardless of architecture choice.

3 Settled Design Properties

The following properties are confirmed by the ETH/USDC 30bps historical backtest (41 days, 600 positions, 2025-12-05 to 2026-01-14) and hold independent of implementation details.

3.1 Backtest-Confirmed Properties

1. **Perpetual.** No fixed expiry. The instrument settles via a funding rate mechanism at liquidity events, following the framework of Angeris et al. [AECB22].
2. **Jump-adjusted funding.** Δ^+ can jump discontinuously when large positions mint or burn, changing N and Θ abruptly. The backtest confirms concentration jumps are rare but large (1 trigger day in 41 days, Δ^+: $0 \to 0.158$). The funding rate accounts for this jump risk per the Primer on Perpetuals.
3. **Discrete funding at liquidity events.** Funding is exchanged when `afterSwap`, `afterAddLiquidity`, or `afterRemoveLiquidity` fires in the underlying pool—matching the FCI update frequency. No keeper or per-block updates required.
4. **USDC numeraire.** Insurance is denominated in USDC (the pool's fee token).
5. **Oracle-assisted framework.** The FCI hook provides $p_{\text{index}} = \Delta^+/(1-\Delta^+)$ as the index price. Following Angeris et al. [AEC21], the oracle-assisted replication framework applies because the hook's own state (A_T, Θ, N) provides the index endogenously.

6. **Monotone payoff.** The protection value is monotone non-decreasing in $p = \Delta^+/(1-\Delta^+)$. The inverted-U from econometrics calibrates *demand* (where PLPs buy protection), not payoff shape.

3.2 Optimal Payoff Formula: Power-Squared Exit Payoff

The backtest compared seven candidate payoff formulas across the ETH/USDC 30bps position dataset:

Mode	Formula	% Better Off	Mean HV ($)
Trigger 1/N (INS-05)	payout $= R/N_{\text{insured}}$	0.0%	−107.67
Trigger premium-prop	payout $\propto$ premium$_i$	2.7%	−95.16
Exit $\times\ \Delta^+$	payout $= \gamma \cdot f \cdot \Delta^+_{\max}$	4.2%	−88.03
Exit $\times\ p$	payout $= \gamma \cdot f \cdot p_{\max}$	8.1%	−71.22
Exit $\times\ p^2$ ($\alpha = 2$)	**payout = premium** $\times\ ((p_{\max}/p^*)^2 - 1)$	**18.8%**	**+23.14**
Exit $\times\ p^3$ ($\alpha = 3$)	payout = premium $\times\ ((p_{\max}/p^*)^3 - 1)$	12.1%	−14.52

(All results at $R_0 = \$200$K seed capital, $\gamma = 3.30\%$, $\Delta^* = 0.09$.)

Definition 3.1 (Power-Squared Exit Payoff)**.** *At position exit (`afterRemoveLiquidity`), PLP i receives:*

$$\textit{payout}_i = \textit{premium}_i \times \left[\left(\frac{\max_t\ p(t)}{p^*}\right)^2 - 1\right]^+ \tag{11}$$

where $\textit{premium}_i = \gamma \times \textit{lifetime_fees}_i$, $p(t) = \Delta^+(t)/(1-\Delta^+(t))$ is the concentration price at time t, and $\max_t$ is taken over the position's lifetime.

3.3 Why Power-Squared Dominates

1. **No moral hazard.** The payout depends on the *worst* (highest) concentration experienced during the position's lifetime. Staying in the pool during concentration spikes *increases* the payout. Exiting early reduces it. This is the opposite of the premium-proportional moral hazard.

2. **Tail sensitivity.** The $\alpha = 2$ exponent is the sweet spot for rare severe events at $\Delta^* = 0.09$: sufficient convexity to reward extreme concentration without draining the reserve at moderate Δ^+. Higher α (e.g., $\alpha = 3$) over-concentrates payouts on the single worst event; lower α (e.g., $\alpha = 1$) provides insufficient tail compensation.

3. **Incentive alignment.** Backtest confirms positive correlation between position lifetime and hedge value: longer-lived positions benefit more than short-lived ones (the correct incentive structure for insurance).

4. **Bootstrapping resolution.** Equal pro-rata (INS-05) yields 0% of positions better off at any γ due to the bootstrapping problem: the mutual reserve is too small at trigger time. The exit payoff sidesteps bootstrapping entirely by settling at exit, not at trigger.

4 Architectural Design

The architecture is a **Lendgine-style CFMM** following Leifke & Scott [Lei24], adapted from ETH price to fee concentration price $p = \Delta^+/(1-\Delta^+)$.

4.1 Participants and Token Mapping

Definition 4.1 (Insurance CFMM Roles).

- ***Underwriter (LP of insurance CFMM):*** *External actor who deposits USDC collateral into a tick range $[p_l, p_u]$. Earns borrow rate (premium stream). Commits to $-p^2$ payoff: loses collateral quadratically as concentration rises.*
- ***PLP (Borrower):*** *Uniswap V4 LP who borrows the insurance CFMM's LP share. Receives $+p^2$ protection payoff. Pays borrow rate from V4 accrued fees.*
- ***Token X (virtual):*** *Concentration exposure token—not a real ERC-20 but an internal accounting unit representing units of concentration risk priced by the FCI oracle.*
- ***Token Y (real):*** *USDC—underwriter's collateral, PLP's premium currency.*

4.2 LP Payoff (Underwriter)

Within tick range $[p_l, p_u]$, the underwriter's portfolio value per unit liquidity:

$$V(p) = \begin{cases} \dfrac{p_u^2 - p_l^2}{p_l^2} & p \le p_l \\ \dfrac{p_u^2 - p^2}{p_l^2} & p_l \le p \le p_u \\ 0 & p \ge p_u \end{cases} \tag{12}$$

Proposition 4.2 (Concavity). *$V(p)$ is concave: $V''(p) = -2/p_l^2 < 0$ in $[p_l, p_u]$.*

4.3 Borrower Payoff (PLP)

The PLP borrows the LP share and receives the convex complement:

$$N(p) = V(p_l) - V(p) = \begin{cases} 0 & p \le p_l \\ \left(\frac{p}{p_l}\right)^2 - 1 & p_l \le p \le p_u \\ \left(\frac{p_u}{p_l}\right)^2 - 1 & p \ge p_u \end{cases} \tag{13}$$

This matches the backtest-confirmed payoff (theorem 3.1) with $p^* = p_l$ (the lowest active underwriter tick) and cap at p_u.

4.4 Market-Implied Strike

The strike p^* is *not* a protocol parameter. It emerges from the market:

$$p^* = \min\{p_l : \exists \text{ underwriter with liquidity at } p_l\} \tag{14}$$

Underwriters express their view on where concentration becomes dangerous by choosing their tick range $[p_l, p_u]$. The aggregate liquidity profile across ticks is the insurance pricing curve. As market conditions change, underwriters reposition, and p^* adjusts naturally—no governance required.

4.5 Connection to PowerMaker

The architecture maps directly onto the PowerMaker [Lei24] Lendgine construction:

PowerMaker	ThetaSwap Insurance	Role
ETH price p	Concentration price $p = \Delta^+/(1-\Delta^+)$	Underlying
LP (deposits into CFMM)	Underwriter (deposits USDC)	Sells convexity
Borrower (borrows LP share)	PLP (borrows LP share)	Buys convexity
$V(p) = 2pk - p^2$	$V(p) = (p_u^2 - p^2)/p_l^2$	LP payoff
$\varphi = R_1 - (k - \frac{1}{2}R_2)^2$	$\psi = y - \frac{p_l^2}{4}u$	Trading function
JumpRate borrow curve	JumpRate + γ base	Borrow rate

5 Methodology and Data

This section presents the full econometric identification of insurance demand from fee concentration risk.

The primary finding is that, for the ETH/USDC 30bps pool we study, there exists a concentration threshold ($\Delta^* \approx 0.09$) beyond which JIT arrival

crowds out passive liquidity providers. In pools where trading volume is inelastic to liquidity depth [CJZ24], JIT positions dilute passive LPs' effective fee rate without generating proportional volume—shifting concentration from a signal of pool health to a driver of LP exit. This regime shift is the empirical foundation of insurance demand.

The goal is to identify precisely where this transition occurs and to translate the excess exit risk into a premium for fee concentration derivatives. This motivates two questions:

1. Does fee concentration risk (A_T) drive passive LP exits in Uniswap V3?

2. At what concentration threshold does the exit effect activate, and can it be priced as an insurance premium?

5.0.1 Unit of Observation

Position-day panel. One observation per position per day over a 41-day window (2025-12-05 to 2026-01-14). Binary outcome: $\text{exit}_{i,t} \in \{0, 1\}$. Treatment: fee concentration deviation from the competitive null, measured at the pool-day level with lag.

5.0.2 Outcome Variable

Binary exit indicator:

$$Y_{i,t} = \begin{cases} 1 & \text{if position } i \text{ burns on day } t \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

Panel size: 3,365 position-day observations, 597 exit events, 40 cluster days, from 600 positions in the ETH/USDC 30bps pool.

5.1 Fee Concentration Index (Econometric Definition)

Definition 5.1 (Fee Concentration Index)**.** *For a pool with N active positions on day t, the fee concentration index is:*

$$A_T = \sqrt{\sum_{k=1}^{N} \theta_k \, x_k^2} \tag{16}$$

where $x_k = fee_k / \sum_j fee_j$ is position k's fee share and $\theta_k = 1/\ell_k$ is its turnover rate.

A_T aggregates two dimensions of competitive pressure: *who earns the fees* (x_k) and *how fast positions turn over* (θ_k). High A_T means a few short-lived positions capture most fee revenue.

5.1.1 The Equal-Share Null

Definition 5.2 (Ma–Crapis Null)**.** *Under the Ma & Crapis (2024) symmetric equilibrium with N identical LPs, each earns equal fee share $x_k = 1/N$:*

$$A_T^{1/N} = \sqrt{\frac{\sum_k \theta_k}{N^2}} \tag{17}$$

This is the competitive baseline—the value A_T takes when fee concentration is zero.

5.1.2 Concentration Deviation

Definition 5.3 (Concentration Deviation)**.** *The treatment variable is the excess concentration above the null:*

$$\Delta_t \equiv A_{T,t}^{real} - A_{T,t}^{1/N} \tag{18}$$

*computed from the **same** set of positions on each day. $\Delta_t > 0$ indicates concentration above the competitive equilibrium; $\Delta_t < 0$ indicates more equal fee distribution than the symmetric benchmark.*

5.1.3 Empirical Properties of Δ_t

Data source: Dune query 6783604, joining `Collect`, `Burn`, and `Mint` events for the ETH/USDC 30bps pool (`0x8ad599c3...eb6e6d8`). Fee amounts isolated by subtracting Burn principal from Collect amounts. Token ordering: token0 = USDC (6 dec), token1 = WETH (18 dec).

Statistic	Value
Days with $\Delta > 0$	26/41 (63%)
Mean $A_T^{\text{real}}/A_T^{1/N}$	2.65×
Median deviation	+0.0004
Range	[−0.032, +0.158]
Positions per day	13–65

The real A_T exceeds the null on most days, confirming empirically that fee shares are asymmetric—consistent with Aquilina et al. finding that 7% of LP addresses capture 80% of fees.

5.2 Theoretical Foundations

Five papers independently validate the components of A_T and the exit mechanism.

5.2.1 Capponi & Zhu (2024): Optimal Exit Threshold

"Optimal Exiting for Liquidity Provision in Constant Function Market Makers."

The LP solves an optimal stopping problem with exit thresholds p_U^* and p_L^*. The key comparative static:

$$\frac{\partial p_U^*}{\partial \phi} > 0 \tag{19}$$

A higher effective fee rate $\hat{\phi}$ raises the exit threshold (LP stays longer). When A_T rises, passive LP's effective fee rate $\hat{\phi}_i = x_i \cdot \phi_{\text{pool}}$ drops because $x_i < 1/N$. Lower $\hat{\phi} \Rightarrow$ lower $p_U^* \Rightarrow$ earlier exit.

Implication for A_T: A_T enters the exit decision through $\hat{\phi}$. This predicts $\beta_1 > 0$ for sufficiently high concentration.

5.2.2 Capponi, Jia & Zhu (2024): JIT Fee Dilution

"The Paradox of Just-in-Time Liquidity in Decentralized Exchanges."

JIT liquidity creates fee share asymmetry: the passive LP's share is $d_P/(d_P + d_J)$ where d_J is JIT depth. This is exactly the x_k component of A_T. JIT entry mechanically raises A_T by concentrating fees in short-lived positions.

Implication for identification: JIT activity is a natural instrument for A_T. Our JIT control analysis (32 JIT positions, 5.3% share) shows JIT count has an independent positive effect on exits ($\beta_{\text{JIT}} = 0.177$, $p = 0.031$), but does not explain the A_T effect—confirming that A_T captures concentration *beyond* JIT.

5.2.3 Ma & Crapis (2024): The Competitive Null

"Decentralized Exchange Design: Permissionless Liquidity Provision."

N symmetric LPs in free-entry equilibrium: each earns $\Pi_i = (1/N^2) f B_D$. The equal-share outcome $x_k = 1/N$ is the **null hypothesis** of A_T. Our deviation $\Delta_t = A_T^{\text{real}} - A_T^{1/N}$ tests departure from this equilibrium directly.

Implication for the model: $\Delta_t = 0$ means the pool operates at the competitive equilibrium. $\Delta_t > 0$ means fee concentration exceeds what symmetric competition predicts. The deviation is the correct treatment variable—not A_T in levels.

5.2.4 Aquilina, Foley, Gambacorta & Krekel (2024): Empirical Validation

"Decentralised Dealers? Examining Liquidity Provision in Decentralised Exchanges." BIS Working Paper No. 1227.

Sophisticated LPs (7% of addresses) provide 65–85% of liquidity, use tick ranges 3.4× tighter, hold positions 7.5× shorter, and capture 75–90% of fees. This validates that x_k in A_T is highly skewed, not symmetric.

Their profitability decomposition $R_{\text{net}} = \text{FeeYield} + \text{IL} - \text{Gas}$ shows that passive LPs earn negative returns when sophisticated LPs dominate—precisely when Δ_t is large.

5.2.5 Bichuch & Feinstein (2024): Fee-as-Derivative Pricing

"The Price of Liquidity: Implied Volatility of AMM Fees."

The LP value function is an optimal stopping problem with implied fee rate $F(p_x, p_y) = p_y \cdot \ell(p_x/p_y)$ where ℓ is instantaneous LVR. Theorem 4.2: a risk-neutral LP is indifferent across all stopping times. This means *any* deterministic exit driver must come from fee rate heterogeneity across LPs—exactly what A_T measures.

Their fixed-for-floating fee swap (Corollary 5.5) is structurally identical to ThetaSwap's insurance product: the LP exchanges volatile realized fees for a fixed premium.

Implication: A_T measures the *distribution* of the pool fee rate across LPs. Bichuch & Feinstein's framework prices the pool-level fee; A_T determines who receives it.

5.3 Economic Model

5.3.1 Environment and Actors

Single Uniswap V3 pool (ETH/USDC 30bps). Two agent types:

1. **Passive LP (PLP):** Commits liquidity for extended periods. Observes pool state (reserves, price, accumulated fees). Does *not* observe A_T directly but reacts to its symptoms (lower fee revenue). Decision: when to exit.

2. **JIT/Sophisticated LP:** Not explicitly modeled as strategic. Enters as a latent force that raises A_T by capturing disproportionate fee shares. Characterized by short blocklife and tight tick ranges (Aquilina et al.).

5.3.2 Equilibrium Concept

Competitive (price-taking) per Capponi & Zhu (2024). The PLP takes fee rate, price process, and concentration risk as given and solves an optimal stopping problem.

Assumption (Price-Taking LP). *The PLP solves:*

$$\max_{\tau} \mathbb{E}\left[\int_0^{\tau} e^{-rt}\hat{\phi}_i \, dF_t - dL_t \;\middle|\; \mathcal{F}_0\right] \tag{20}$$

where $\hat{\phi}_i = x_i \cdot \phi_{\text{pool}}$ is the LP's effective fee rate given fee share x_i, F_t is cumulative fee revenue, and L_t is impermanent loss.

5.4 Stochastic Model

5.4.1 Unobserved Heterogeneity (η)

1. **LP risk aversion** (η_i^{risk}): heterogeneous preferences. Some PLPs tolerate more concentration. NFT Position Manager masks identity.
2. **Outside options** (η_i^{outside}): competing pool returns, staking yields, CEX opportunities.

Bias direction: **toward zero** (conservative). Risk-averse LPs avoid entering during high-A_T periods *and* exit faster, attenuating β.

5.4.2 Agent Uncertainty

Future price process only ($u_t \sim$ GBM, per Capponi & Zhu). Drives IL realization.

5.4.3 Implied Error Structure

$$\varepsilon_i = \underbrace{\eta_i^{\text{risk}} + \eta_i^{\text{outside}}}_{\text{unobserved heterogeneity}} + \underbrace{\nu_i}_{\text{measurement error}} \tag{21}$$

5.5 Estimation Strategy

5.5.1 Model Progression

We estimate three nested specifications to motivate the quadratic model.

Specification 1: Linear (Point Lag).

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 A_{T,t-\ell} + \beta_2 \,\text{IL}_t + \beta_3 \log(\text{age}_{i,t})\big) \tag{22}$$

Result 1 (Linear A_T). $\beta_1 = -3.93$, cluster $p = 0.072$. Marginally significant but **wrong sign**—higher concentration associated with fewer exits. Robust across lags 1–7 (all negative, 4/5 significant).

Specification 2: Linear Deviation from $1/N$.

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \Delta_{t-\ell} + \beta_2 \,\text{IL}_t + \beta_3 \log(\text{age}_{i,t})\big) \tag{23}$$

where $\Delta_t = A_T^{\text{real}} - A_T^{1/N}$ from the same position set.

Result 2 (Linear Deviation). $\beta_1 = -5.44$, cluster $p = 0.109$. Still negative. But **dose-response analysis reveals non-monotonicity**:

Quartile	Mean Δ	Exit Rate	Pattern
Q1 (below null)	-0.009	17.75%	baseline
Q2 (at null)	-0.000	17.87%	baseline
Q3 (mild conc.)	**+0.002**	**20.56%**	**exits increase**
Q4 (extreme conc.)	$+0.025$	14.37%	exits decrease

Q3 shows the predicted Capponi & Zhu pattern. Q4 reverses—motivating the quadratic specification.

Specification 3: Quadratic Deviation (Primary Model).

$$\boxed{P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \Delta_{t-\ell} + \beta_2 \Delta_{t-\ell}^2 + \beta_3 \,\text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)} \quad (24)$$

This is the **primary specification**. The quadratic term captures the inverted-U: shelter at low concentration, Capponi exit at high concentration.

5.5.2 Estimation Method

Logit MLE via Newton–Raphson (IRLS) with **day-clustered sandwich standard errors** (41 clusters):

$$\widehat{\text{Var}}(\hat{\beta}) = H^{-1} \left(\sum_{g=1}^{G} S_g S_g' \right) H^{-1} \cdot \frac{G}{G-1} \cdot \frac{n}{n-k} \quad (25)$$

where $S_g = \sum_{i \in g} (y_i - \hat{p}_i) x_i$ is the cluster-level score and H is the Hessian.

Implemented in JAX 0.9.1 CPU. Code: `econometrics/hazard.py`.

5.6 Results

5.6.1 Main Result: Quadratic Deviation Model

Lag	$\hat{\beta}_1$ (linear)	p	$\hat{\beta}_2$ (quadratic)	p	Turning point
1	**−23.18**	**0.012**	**+129.20**	**0.030**	**0.090**
2	−43.42	*0.016*	+226.92	*0.065*	*0.096*
3	**−32.44**	**0.001**	**+205.34**	**0.004**	**0.079**
5	+3.81	0.830	−76.31	0.470	—
7	−5.55	0.730	−16.46	0.900	—

Both terms of the quadratic deviation model are statistically significant at the economically relevant lag horizon (1–3 days). The turning point clusters at $\Delta^* \approx 0.08$–0.10. The signal dissipates at lags 5–7, consistent with PLPs responding to recent concentration shocks, not stale ones.

5.6.2 Full Lag-1 Coefficients

Parameter	Estimate	Cluster SE	Interpretation
β_0 (intercept)	varies	—	baseline exit probability
β_1 (Δ)	-23.18	9.21	shelter below turning point
β_2 (Δ^2)	$+129.20$	59.69	exit above turning point
β_3 (IL)	$+13.52$	—	higher IL $\rightarrow$ more exits
β_4 (log age)	-0.42	—	older positions less likely to exit

n = 3,365; exits = 597; clusters = 40; pseudo-R^2 = 0.350; mean $P(\text{exit}) = 0.177$.

5.7 Economic Interpretation

5.7.1 The Inverted-U: Two Regimes

The quadratic model identifies two regimes separated by the turning point $\Delta^* \approx 0.09$:

1. **Shelter regime** ($\Delta < \Delta^*$): Mild fee concentration correlates with high trading volume. All LPs benefit from elevated fee revenue, even if shares are unequal. The marginal effect $\partial P/\partial\Delta < 0$: more concentration, fewer exits.

2. **Capponi regime** ($\Delta > \Delta^*$): Extreme concentration means passive LPs' effective fee rate $\hat{\phi}_i$ drops below their exit threshold. Per Capponi & Zhu: $\partial p_U^*/\partial\phi > 0$, so lower $\phi \Rightarrow$ lower $p_U^* \Rightarrow$ earlier exit. The marginal effect flips: $\partial P/\partial\Delta > 0$.

5.7.2 Marginal Effect at the Turning Point

The marginal effect of the quadratic model is:

$$\frac{\partial P(\text{exit})}{\partial\Delta} = (\beta_1 + 2\beta_2\Delta) \cdot \bar{P}(1 - \bar{P}) \tag{26}$$

At the turning point $\Delta^* = -\beta_1/(2\beta_2) = 0.090$:

$$\left.\frac{\partial P}{\partial\Delta}\right|_{\Delta^*} = 0$$

For $\Delta = 0.15$ (well into the Capponi regime):

$$\beta_1 + 2\beta_2(0.15) = -23.18 + 2(129.20)(0.15) = 15.58 \tag{27}$$

$$\frac{\partial P}{\partial\Delta} = 15.58 \times 0.177 \times 0.823 = 2.27 \tag{28}$$

A 0.01 increase in Δ at this level raises exit probability by $\sim$2.3 percentage points.

5.7.3 Insurance Pricing Translation

For a passive LP experiencing $\Delta = 0.15$ (concentration 15% above the competitive null):

Quantity	Value
$\Delta P(\text{exit})$ per 0.01 Δ	+2.3 pp
Average remaining position hours	48
Hours of fee revenue at risk	$0.023 \times 48 = 1.1$ hrs
Fee revenue per hour (pool avg)	\$100
Implied premium per position	\$110

This is the maximum willingness-to-pay for ThetaSwap's insurance: a PLP would pay up to \$110 to eliminate the excess exit risk from concentration above the turning point.

5.8 Specification Tests

5.8.1 Test 1: Sign Restriction (Capponi & Zhu Prediction)

Hypothesis 1. $\beta_2 > 0$: the quadratic term is positive, indicating that extreme concentration drives exits.

Result: PASS. $\hat{\beta}_2 = +129.20$, $p = 0.030$ at lag 1; $+205.34$, $p = 0.004$ at lag 3. The Capponi exit mechanism is confirmed in the high-concentration regime.

5.8.2 Test 2: Dose-Response Monotonicity

Hypothesis 2. Exit rates increase with Δ in the upper quartiles.

Result: PASS (inverted-U). Q1–Q2 stable (17.8%), Q3 peak (20.6%), Q4 drops (14.4%). The non-monotonicity motivates the quadratic and is captured by the model.

5.8.3 Test 3: Lag Robustness

Hypothesis 3. Both β_1 and β_2 are significant for at least 3 of 5 lag windows.

Result: PASS. Lags 1, 2, 3 all have significant quadratic terms ($p < 0.07$). The turning point is stable at 0.08–0.10. Signal dissipates at lags 5–7, consistent with short-memory PLP reaction.

5.8.4 Test 4: IL Orthogonality

Hypothesis 4. β_1, β_2 persist with and without the IL control.

Result: PASS. IL coefficient ($\beta_3 = 13.52$) is positive (higher IL $\rightarrow$ more exits) but does not absorb the concentration effect. The quadratic structure is present in the A_T-only model as well.

5.8.5 Test 5: JIT Confounding Control

Hypothesis 5. The concentration effect is not an artifact of JIT activity.

Result: PASS. Adding a JIT proxy (positions with blocklife < 100 blocks) as a control:

- JIT count has independent positive effect on exits ($\beta_{\text{JIT}} = 0.177$, $p = 0.031$)
- The A_T effect becomes *more* significant ($p : 0.072 \to 0.022$) with JIT controlled
- Confirms: A_T captures concentration *beyond* JIT

5.8.6 Test 6: Alternative Treatment Timing

Three treatment constructions tested:

Treatment	β_1 (linear)	p	Finding
Point A_T (real, lag 1)	-3.93	0.072	Negative (shelter)
Lifetime mean A_T	-7.60	0.444	Negative, insignificant
Deviation from $1/N$	-5.44	0.109	Negative (Q3 reveals non-linearity)
Quadratic deviation	**See Result 3**		**Inverted-U confirmed**

5.9 Robustness: Five Linear Specifications

Before arriving at the quadratic model, we tested five linear specifications that progressively refined the treatment variable. All failed to produce a positive β_1, but each contributed to understanding the data-generating process.

5.9.1 Specification A: Point Real A_T

Treatment $= A^{\text{real}}_{T,t-\ell}$ (single day's concentration).

Lag	$\hat{\beta}_1$	Cluster SE	p	Sig
1	-3.928	2.186	0.072	*
2	-8.783	2.227	0.0001	***
3	-0.414	1.131	0.714	
5	-6.523	1.650	0.0001	***
7	-5.625	2.261	0.013	**

Finding: Consistently negative. A naive interpretation is that concentration *protects* LPs. But the proxy A_T ($x_k = 1/N$) showed the same pattern ($\beta_1 = -4.90$, $p = 0.15$), suggesting the level of A_T confounds pool health (volume) with concentration pressure.

5.9.2 Specification B: Lifetime Mean A_T

Treatment = mean A_T^{real} over $[\text{mint}_i, t - \ell]$. Motivated by the hypothesis that passive LPs react to accumulated exposure, not point shocks.

Lag	$\hat{\beta}_1$	Cluster SE	p	Sig
1	-7.595	9.913	0.444	
3	-11.603	7.959	0.145	
5	-14.977	8.021	0.062	*
7	-3.042	6.829	0.656	

Finding: Amplifies the negative effect but with much larger SEs (reduced variation in running means). Marginal significance only at lag 5.

5.9.3 Specification C: Deviation from $1/N$ (Wrong Null)

First attempt at the deviation treatment, using the old proxy A_T from a *different* position set (Q4 burns) as the null. The real and null A_T were computed from different populations—an apples-to-oranges comparison.

Finding: Real A_T was 10–100× smaller than the proxy null (different position sets, different blocklives). Deviation was negative for 40/41 days. Only lag-1 showed positive β_1 (+1.92, $p = 0.43$). **Discarded** after identifying the data mismatch.

5.9.4 Specification D: Corrected Deviation (Same Position Set)

Null A_T recomputed from the *same* Dune query (same positions, same blocklives). Now 26/41 days have $\Delta > 0$, mean ratio = 2.65×.

Finding: Still negative β_1 in linear model, but dose-response revealed the Q3 peak (20.56%) and Q4 dip (14.37%) that motivated the quadratic specification.

5.9.5 Specification E: JIT-Controlled Linear

Added daily JIT count (positions with blocklife < 100 blocks) as control.

Finding: β_1 became *more* negative and more significant ($p : 0.072 \to 0.022$). JIT itself predicts exits but does not confound the A_T effect.

5.10 Literature Alignment

Each component of the quadratic deviation model is validated by at least one paper:

Paper	Component Validated	Connection to Our Result
Capponi & Zhu (2024)	Exit threshold: $\partial p_U^*/\partial\phi > 0$	Confirmed in the Capponi regime ($\Delta > 0.09$): $\beta_2 > 0$ means extreme concentration drives exits
Capponi, Jia & Zhu (2024)	x_k captures JIT fee dilution	JIT control shows A_T effect is independent of JIT confounding; x_k reflects broader concentration than just JIT
Ma & Crapis (2024)	$x_k = 1/N$ is the competitive null	Deviation $\Delta = A_T^{\text{real}} - A_T^{1/N}$ is the correct treatment; levels confound pool health
Aquilina et al. (2024)	7% of LPs capture 80% of fees	Real A_T is 2.65× the null on average, confirming massive fee share asymmetry
Bichuch & Feinstein (2024)	Risk-neutral LP indifferent on stopping times	Any exit driver must come from fee *heterogeneity* across LPs—exactly what Δ measures

5.11 Connection to ThetaSwap

1. **Insurance trigger:** The turning point $\Delta^* \approx 0.09$ defines when insurance becomes valuable. When concentration exceeds 9% above the Ma–Crapis equilibrium, passive LP exit risk increases. ThetaSwap monitors Δ_t in real time and activates payouts when $\Delta > \Delta^*$.

2. **Premium calibration:** The quadratic marginal effect at any Δ translates directly to an insurance premium via eq. (26). At $\Delta = 0.15$: \$110 per position per event.

3. **Distinct from IL hedging:** The IL coefficient (β_3) is positive but does not absorb the concentration effect. A_T captures a risk orthogonal to price-movement risk, confirming that ThetaSwap addresses a gap unmet by options or IL protection products.

4. **Product structure:** Per Bichuch & Feinstein's fixed-for-floating fee swap framework, ThetaSwap is a derivative that pays the PLP when realized fee share x_i falls below $1/N$ by more than Δ^*. The premium is the implied volatility of fee shares.

5. **Market sizing:** 63% of days in our sample have $\Delta > 0$. Of these, days exceeding Δ^* represent the insurance-triggering events. Aggregating

the implied premium across 600 positions gives the minimum viable insurance pool size.

5.12 Data and Implementation

Query	Description	Rows	Cos
Q4v2	Position-level burns, blocklife	600	0.18
Q5	Daily IL proxy	41	0.14
Q6	Per-position Collect fees + blocklife (real A_T and null A_T per day)	41 (aggregated)	3.3

Total cost: ∼3.6 credits.

Pool: ETH/USDC 30bps (`0x8ad599c3a0ff1de082011efddc58f1908eb6e6d8`).
Window: 2025-12-05 to 2026-01-14 (41 days).
Implementation: Python 3.11, JAX 0.9.1 CPU, `@functional-python` patterns (frozen dataclasses, pure functions). Code in `econometrics/{types,data,ingest,hazard}`
52 unit tests passing.

Key functions:

- `build_exit_panel_deviation()`: constructs position-day panel with Δ as treatment
- `logit_mle_quadratic()`: Newton–Raphson logit with $\Delta + \Delta^2$ and clustered SEs
- `marginal_effect_at_means()`: translates β to insurance premium

5.13 Econometric References

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6 Trading Function

6.1 Reserve Derivation

Starting from the LP payoff $V(p) = (p_u^2 - p^2)/p_l^2$ for $p \in [p_l, p_u]$, we derive reserves via the Angeris replication theorem [AEC21].

Definition 6.1 (Reserves)**.** *The risky (virtual concentration) and numeraire (USDC) reserves per unit liquidity:*

$$x(p) = -V'(p) = \frac{2p}{p_l^2} \tag{29}$$

$$y(p) = V(p) + p \cdot x(p) = \frac{p_u^2 + p^2}{p_l^2} \tag{30}$$

for $p \in [p_l, p_u]$, with boundary extensions: $x = 2p_l/p_l^2$ and $y = (p_u^2 + p_l^2)/p_l^2$ for $p \leq p_l$; $x = 2p_u/p_l^2$ and $y = 2p_u^2/p_l^2$ for $p \geq p_u$.

Proposition 6.2 (Reserve Verification)**.** *$V(p) = y(p) - p \cdot x(p)$ holds exactly:*

$$\frac{p_u^2 + p^2}{p_l^2} - p \cdot \frac{2p}{p_l^2} = \frac{p_u^2 + p^2 - 2p^2}{p_l^2} = \frac{p_u^2 - p^2}{p_l^2} = V(p) \qquad \checkmark$$

6.2 Coordinate Transform: $u = x^2$

The natural trading function $y = p_u^2/p_l^2 + (p_l^2/4)x^2$ is quadratic in x. This creates a 2-homogeneous invariant, incompatible with O(1) tick-based liquidity aggregation (Uniswap V3-style $L_{\text{act}} \mathrel{+}= \Delta L_{\text{net}}$ at tick crossings).

Definition 6.3 (Linearization)**.** *Define the transformed risky reserve $u = x^2$. Then:*

$$\boxed{\psi(u, y) = y - \frac{p_l^2}{4}\, u} \tag{31}$$

with invariant constant $C_1 = p_u^2/p_l^2$ per unit liquidity.

Theorem 6.4 (Linearity and 1-Homogeneity)**.** *$\psi(u, y)$ is linear in (u, y). With liquidity L:*

$$u_{total} = L \cdot \tilde{u}(p), \quad y_{total} = L \cdot \tilde{y}(p)$$

where $\tilde{u}(p) = 4p^2/p_l^4$ and $\tilde{y}(p) = (p_u^2 + p^2)/p_l^2$ are per-unit-liquidity reserves. The invariant scales as:

$$\psi(u_{total}, y_{total}) = L \cdot C_1$$

which is 1-homogeneous in L.

Proof. $\psi(L\tilde{u}, L\tilde{y}) = L\tilde{y} - (p_l^2/4) \cdot L\tilde{u} = L(\tilde{y} - (p_l^2/4)\tilde{u}) = L \cdot C_1$. □

Remark 6.5 (Concavity)**.** *The Hessian of ψ has eigenvalues $-p_l^2/2$ and 0, both ≤ 0. Hence ψ is concave, and the trading set $S = \{(u, y) : \psi(u, y) \geq LC_1\}$ is convex (Geometry paper [ACD+23], Theorem 1).*

6.3 Spot Price

Definition 6.6 (Spot Price from Reserves)**.**

$$p = \frac{p_l^2}{2}\sqrt{\frac{u}{L}} = \frac{p_l^2}{2} \cdot \frac{x}{L} \quad \textit{(in } x \textit{ space: } p = p_l^2 x/(2L)\textit{)} \tag{32}$$

where the second equality uses $x = \sqrt{u}$.

Verification. From the trading function, $-\partial\psi/\partial u / \partial\psi/\partial y = -(-p_l^2/4)/1 = p_l^2/4$. This is the marginal price in (u, y) space. In (x, y) space, $du = 2x\,dx$, so the marginal price is $dy/dx = (p_l^2/4) \cdot 2x = p_l^2 x/2 = p$, matching the reserve derivation $x(p) = 2p/p_l^2 \Leftrightarrow p = p_l^2 x/2$. □

6.4 Swap Mechanics

A swap in the insurance CFMM operates in (u, y) space internally, with $x \leftrightarrow u$ conversion at the boundary.

Definition 6.7 (Swap)**.** *Trader sells Δx units of concentration exposure:*

$$\Delta u = (x + \Delta x)^2 - x^2 = 2x\,\Delta x + (\Delta x)^2 \tag{33}$$

$$\Delta y = \frac{p_l^2}{4}\,\Delta u \tag{34}$$

The effective price $\Delta y/\Delta x = (p_l^2/4)(2x + \Delta x)$ depends on the current state x and trade size Δx, providing natural slippage despite the internal invariant being linear.

Remark 6.8 (O(1) Complexity)**.** *Within a tick: the swap is a single multiplication (eq. (34)) plus one square root for $x = \sqrt{u}$ conversion (`FixedPointMathLib.sqrt`). Tick crossing: L_{act} += ΔL_{net}, a single addition. Total per-swap cost: O(ticks crossed), same as Uniswap V3.*

6.5 Liquidity Provisioning

Definition 6.9 (Single-Sided USDC Deposit)**.** *An underwriter deposits Δy USDC at current price p:*

$$\Delta L = \frac{\Delta y \cdot p_l^2}{p_u^2 + p^2} \tag{35}$$

$$\Delta u = \Delta L \cdot \frac{4p^2}{p_l^4} = \frac{4p^2 \cdot \Delta y}{p_l^2(p_u^2 + p^2)} \tag{36}$$

The virtual Δu is computed, not deposited. Only USDC enters the contract.

Proposition 6.10 (Invariant Preservation on Mint)**.** *After minting:* $\psi(u + \Delta u, y + \Delta y) = (L + \Delta L) \cdot C_1$.

Proof. $\psi(u + \Delta u, y + \Delta y) = (y + \Delta y) - (p_l^2/4)(u + \Delta u) = \psi(u, y) + \Delta y - (p_l^2/4)\Delta u = LC_1 + \Delta L \cdot \tilde{y} - (p_l^2/4)\Delta L \cdot \tilde{u} = LC_1 + \Delta L \cdot C_1 = (L + \Delta L)C_1$. □

7 Reserve Analysis

This section verifies the reserve curves derived in section 6 and establishes boundary behavior and numerical properties.

7.1 Reserve Curves in Price Space

Per unit liquidity ($L = 1$), with $p \in [p_l, p_u]$:

	$p = p_l$	$p \in (p_l, p_u)$	$p = p_u$
$x(p) = 2p/p_l^2$	$2/p_l$	$2p/p_l^2$	$2p_u/p_l^2$
$u(p) = x^2 = 4p^2/p_l^4$	$4/p_l^2$	$4p^2/p_l^4$	$4p_u^2/p_l^4$
$y(p) = (p_u^2 + p^2)/p_l^2$	$(p_u^2 + p_l^2)/p_l^2$	$(p_u^2 + p^2)/p_l^2$	$2p_u^2/p_l^2$
$V(p) = (p_u^2 - p^2)/p_l^2$	$(p_u^2 - p_l^2)/p_l^2$	$(p_u^2 - p^2)/p_l^2$	0

7.2 Monotonicity

Proposition 7.1 (Reserve Monotonicity)**.** *Both $x(p)$ and $y(p)$ are strictly increasing in p on $[p_l, p_u]$:*

$$x'(p) = \frac{2}{p_l^2} > 0, \qquad y'(p) = \frac{2p}{p_l^2} > 0 \quad \textit{for } p > 0$$

Remark 7.2. *Both reserves increasing with p is the correct direction under Framing C (section 4): as concentration rises, more virtual exposure is absorbed (x rises) and more numeraire accumulates (y rises) because the underwriter's collateral drains into the payoff liability. The LP value $V(p) = y - px$ decreases, which is the underwriter bearing losses.*

7.3 Collateral Accounting

The underwriter deposits USDC (Token Y). The virtual Token X reserve is an internal accounting variable with no external token.

Definition 7.3 (Collateral Value)**.** *At price p, the underwriter's real (USDC) collateral per unit liquidity equals the LP value:*

$$\textit{collateral}(p) = V(p) = y(p) - p \cdot x(p) = \frac{p_u^2 - p^2}{p_l^2} \tag{37}$$

At $p = p_l$: full collateral $(p_u^2 - p_l^2)/p_l^2$. At $p = p_u$: zero (fully drained). The underwriter cannot lose more than their initial deposit—the CLAMM tick cap p_u enforces bounded loss.

7.4 Numerical Verification

For $\Delta^* = 0.09$, $p_l = 0.0989$, $p_u = 1.0$ ($\Delta^+_{\max} = 0.5$):

p	Δ^+	x	y	V (USDC/unit L)
0.0989 ($= p_l$)	0.09	20.22	103.2	101.2
0.1758	0.149	35.95	105.4	99.04
0.5	0.333	102.2	127.6	76.5
1.0 ($= p_u$)	0.5	204.5	204.5	0

Payoff check at $p = 0.1758$: $N = V(p_l) - V(p) = 101.2 - 99.04 = 2.16$. Formula: $(0.1758/0.0989)^2 - 1 = 3.16 - 1 = 2.16$ ✓.

8 Borrow Rate

The PLP (borrower) pays two separate rates to the underwriter (lender): a unidirectional borrow rate (always PLP → underwriter) and a bidirectional funding rate (section 9). These are economically distinct mechanisms with separate accumulators, following the Primer on Perpetuals [AECB22].

8.1 Borrow Rate: JumpRate Curve

Definition 8.1 (Borrow Rate)**.**

$$r_{borrow}(U_a) = \gamma + m_1 \cdot \min(U_a, \kappa) + m_2 \cdot \max(0,\, U_a - \kappa) \tag{38}$$

where:

- $\gamma > 0$*: base premium (floor), calibrated from econometric WTP;*
- $U_a = L_{borrowed}/L_{total} \in [0, 1]$*: utilization ratio;*
- $\kappa \in (0, 1)$*: target utilization (e.g., $\kappa = 0.8$);*
- $m_1 > 0$*: slope below kink (gentle increase);*
- $m_2 \gg m_1$*: slope above kink (jump multiplier, strong deterrent).*

Proposition 8.2 (Borrow Rate Properties)**.** *(i) $r_{borrow} \geq \gamma > 0$ for all $U_a \geq 0$ (underwriters always earn at least γ).*

(ii) r_{borrow} is continuous and piecewise-linear in U_a.

(iii) r_{borrow} *is non-decreasing in* U_a *(higher utilization* $\rightarrow$ *higher borrow cost).*

(iv) At $U_a = 1$*:* $r_{borrow} = \gamma + m_1\kappa + m_2(1-\kappa)$ *(very expensive, incentivizing new underwriter deposits or PLP exit).*

8.2 Borrow Rate Accrual

Definition 8.3 (Borrow Fee Accumulator)**.** *At each hook callback (afterSwap, afterAddLiquidity, afterRemoveLiquidity), the borrow fee growth accumulator updates:*

$$g^{borrow} \mathrel{+}= r_{borrow}(U_a) \cdot \Delta t \tag{39}$$

where Δt *is the time elapsed since the last callback (in blocks or seconds). This accumulator is global (per insurance pool) and distributed to underwriters pro-rata by liquidity share, using the Synthetix staking pattern [BBJ18].*

8.3 Separate Accounting Principle

Remark 8.4 (Two Accumulators, Two Purposes)**.** *The borrow rate and funding rate (section 9) use separate accumulators:*

- g^{borrow}*: always positive, PLP* $\rightarrow$ *underwriter. Compensates capital lockup and tail risk.*
- $g^{funding}$*: bidirectional. Corrects mark-index mispricing. When positive, PLP pays; when negative, underwriter pays.*

The Primer on Perpetuals [AECB22, Section 3] establishes that conflating these mechanisms breaks funding convergence: a funding term that cannot go negative (because it's floored by the borrow rate) cannot fully correct underpricing.

9 Funding Rate

This instrument is perpetual: it has no fixed expiry but settles at liquidity events. The funding rate maintains mark-index convergence, following the framework of [AECB22]. It is separate from the borrow rate (section 8) and bidirectional.

9.1 Mark and Index Prices

Definition 9.1 (Index Price)**.** *The index price is derived from the FCI hook's co-primary state:*

$$p_{index} = \frac{\Delta^+}{1 - \Delta^+} \tag{40}$$

where $\Delta^+ = \max(0,\, A_T - A_T^{1/N})$ *is the realized concentration deviation from the underlying pool. Computed on the fly from stored* A_T, Θ, N.

Definition 9.2 (Mark Price)**.** *The mark price is the CFMM's implied price from virtual reserves:*

$$p_{mark} = \frac{p_l^2}{2}\sqrt{\frac{u}{L_{act}}} \tag{41}$$

where u *is the current virtual concentration reserve and* L_{act} *is the active liquidity in the current tick range.*

9.2 Discrete Funding at Hook Callbacks

Funding is exchanged at each hook callback (`afterSwap`, `afterAddLiquidity`, `afterRemoveLiquidity`), matching the FCI oracle update frequency. No keeper or per-block updates required.

Definition 9.3 (Funding Payment)**.** *At callback* n*, the funding rate is:*

$$r_{fundingn} = \alpha \cdot \frac{p_{markn} - p_{indexn}}{p_{indexn} + 1} \tag{42}$$

where $\alpha > 0$ *is the sensitivity parameter (set at pool initialization). The denominator* $(p_{index} + 1)$ *normalizes by index level, preventing extreme rates when* p_{index} *is small.*

Remark 9.4 (Bidirectionality)**.**

- $p_{mark} > p_{index}$ *(protection overpriced):* $r_{funding} > 0$*, PLPs pay underwriters. Incentivizes arbitrageurs to sell the perpetual, pushing* p_{mark} *down.*
- $p_{mark} < p_{index}$ *(protection underpriced):* $r_{funding} < 0$*, underwriters pay PLPs. Incentivizes buying the perpetual, pushing* p_{mark} *up.*

This bidirectionality is essential for convergence and is why funding must be in a separate accumulator from the unidirectional borrow rate (section 8).

9.3 Jump-Adjusted Replication

The backtest (ETH/USDC 30bps, 41 days) confirms that Δ^+ exhibits rare, large, discontinuous jumps: $0 \to 0.158$ in a single day. Between callbacks, a single large swap or JIT event can change the FCI dramatically. The funding rate includes a jump premium.

Definition 9.5 (Jump Premium)**.** *When* $|\Delta p_{index}| > \delta_{jump}$ *at callback* n *(where* δ_{jump} *is the jump threshold), the funding rate receives an additive adjustment:*

$$r_{fundingn}^{adj} = r_{fundingn} + \lambda \cdot |\Delta p_{index}| \tag{43}$$

where $\lambda > 0$ *is the jump intensity parameter. This compensates underwriters for unhedgeable gap risk between callbacks.*

Theorem 9.6 (Jump Safety). *When Δ^+ jumps discretely from δ to δ' at a liquidity event (due to abrupt changes in N or Θ), the protection value change is bounded:*

$$|\Delta value| \leq |V(\delta') - V(\delta)| \tag{44}$$

where V is the LP payoff function (eq. (12)*).*

Proof. The LP payoff $V(p)$ is Lipschitz continuous on $[p_l, p_u]$ with $|V'(p)| = 2p/p_l^2 \leq 2p_u/p_l^2$. The value change at a jump equals $V(p(\delta')) - V(p(\delta))$, which is bounded by the payoff difference because V is monotone and no intermediate states are observable between callbacks. The jump-adjusted funding compensates for this discontinuity following [AECB22, Section 3]. □

9.4 Funding Accumulator

Definition 9.7 (Funding Growth). *The global funding growth accumulator:*

$$g^{funding} \mathrel{+}= r_{funding_n}^{adj} \cdot \Delta t_n \tag{45}$$

This accumulator can be positive or negative. Per-position funding owed is computed via the Synthetix difference pattern: $funding_i = L_i \cdot (g^{funding} - g^{funding}_{baseline,i})$.

9.5 Convergence Property

Proposition 9.8 (Mark-Index Convergence). *Under the funding mechanism, $|p_{mark} - p_{index}|$ decreases after each callback (in expectation), provided $\alpha > 0$.*

Proof sketch. When $p_{\text{mark}} > p_{\text{index}}$, PLPs pay positive funding, reducing demand for the perpetual and pushing p_{mark} down. When $p_{\text{mark}} < p_{\text{index}}$, underwriters pay, increasing demand and pushing p_{mark} up. The sensitivity α controls convergence speed. A formal convergence proof under specific assumptions on arrival rates is deferred to the Kontrol verification phase. □

10 Initialization

10.1 Pool Initialization

Definition 10.1 (Insurance Pool Parameters). *An insurance CFMM is initialized per underlying V4 pool with:*

(i) `fciHook`*: address of the deployed FeeConcentrationIndex hook (oracle for p_{index});*

(ii) `poolManager`*: Uniswap V4 PoolManager address;*

(iii) $\gamma \in (0,1)$*: base premium factor (borrow rate floor);*

(iv) $\alpha > 0$*: funding rate sensitivity;*

(v) $\lambda > 0$*: jump intensity parameter;*

(vi) $\kappa \in (0,1)$*: utilization target for JumpRate curve;*

(vii) $m_1, m_2 > 0$*: JumpRate slope parameters* ($m_2 \gg m_1$)*;*

(viii) `tickSpacing`*: tick spacing for the insurance CFMM tick array.*

These parameters are immutable after deployment.

10.2 First Underwriter Deposit

Definition 10.2 (Initialization from Oracle)**.** *The first underwriter deposits* D *USDC into tick range* $[p_l, p_u]$*. The FCI hook provides current* Δ^+*, yielding* $p_0 = \Delta^+/(1-\Delta^+)$*.*

Case 1: $p_0 \le p_l$ *(no active concentration—typical at launch):*

$$L_0 = \frac{D \cdot p_l^2}{p_u^2 + p_l^2} \tag{46}$$

$$u_0 = L_0 \cdot \frac{4p_l^2}{p_l^4} = \frac{4L_0}{p_l^2} \tag{47}$$

$$y_0 = D \tag{48}$$

Spot price initializes at p_l *(the strike boundary).*

Case 2: $p_l < p_0 < p_u$ *(concentration already active):*

$$L_0 = \frac{D \cdot p_l^2}{p_u^2 + p_0^2} \tag{49}$$

$$u_0 = L_0 \cdot \frac{4p_0^2}{p_l^4} \tag{50}$$

$$y_0 = D \tag{51}$$

The underwriter enters mid-range with immediate concentration liability.

Case 3: $p_0 \ge p_u$ *(concentration above cap): Deposit reverts. The underwriter would face instant max-loss.*

Proposition 10.3 (Invariant at Initialization)**.** *In all valid cases:* $\psi(u_0, y_0) = L_0 \cdot p_u^2/p_l^2$*.*

Proof. For Case 1: $\psi = D - (p_l^2/4)(4L_0/p_l^2) = D - L_0 = D - Dp_l^2/(p_u^2+p_l^2) = D \cdot p_u^2/(p_u^2+p_l^2) = L_0 \cdot (p_u^2+p_l^2)/p_l^2 \cdot p_u^2/(p_u^2+p_l^2) = L_0 \cdot p_u^2/p_l^2$. □

10.3 PLP Registration (Borrowing)

Definition 10.4 (PLP Borrows LP Share)**.** *A PLP holding a V4 LP position borrows insurance CFMM LP shares:*

(i) PLP calls `borrow(poolKey, v4PositionId, borrowAmount)`*.*

(ii) The hook verifies the PLP holds a valid V4 position with active fee stream.

(iii) LP shares are transferred from the insurance CFMM to the PLP (reducing $L_{available}$, increasing $L_{borrowed}$).

(iv) The PLP's $\max_t p(t)$ tracker is initialized at current p_{index}.

(v) Utilization U_a increases, raising r_{borrow} for all borrowers.

10.4 PLP Exit (Returning LP Share)

Definition 10.5 (Exit Settlement)**.** *At* `afterRemoveLiquidity` *(or voluntary deregistration):*

(i) Compute $premium_i = \gamma \times lifetime_fees_i + accrued_borrow_rate_i + accrued_funding_i$.

(ii) Compute payout: $payout_i = premium_i \times [(\max_t p(t)/p_l)^2 - 1]^+$.

(iii) The payout is capped at $premium_i \times [(p_u/p_l)^2 - 1]$ (tick range cap).

(iv) LP shares are returned to the insurance CFMM. Utilization U_a decreases.

(v) Net settlement: PLP receives $\max(payout - premium_i, 0)$ from the CFMM reserves. If payout $<$ premium, the difference stays in reserves.

Remark 10.6 (Max-Price Tracking)**.** *Each PLP position stores $p_{\max,i} = \max_{t \in [mint, now]} p_{index}(t)$. Updated at every hook callback: $p_{\max,i} \leftarrow \max(p_{\max,i}, p_{index})$. This is one SSTORE per position per callback (only if $p_{index} > p_{\max,i}$), amortized by checking the condition before writing.*

11 Invariants

This section contains all invariants for the fee concentration insurance system. The invariant set is organized into five groups:

1. CFMM-01 through CFMM-19: Base CFMM invariants (trading set, reserves, fees, ticks).

2. CFMM-20 through CFMM-25: Replication invariants (payoff, reserves, canonical form).
3. CFMM-26 through CFMM-28: Perpetual invariants (funding, jump safety).
4. FCI-01 through FCI-13: Fee concentration index invariants (state, price mapping, perpetual funding).
5. INS-01 through INS-10: Insurance CFMM invariants (borrow, settlement, utilization).

11.1 Base CFMM Invariants (CFMM-01 through CFMM-19)

CFMM-01 (*Trading set convexity*).

$$S = \{(u, y) : \psi(u, y) \geq L \cdot C_1\} \text{ is convex}$$

The trading set is convex. Since $\psi(u, y) = y - (p_l^2/4)u$ is linear (hence concave), the superlevel set is convex (Geometry paper [ACD+23], Theorem 1).

CFMM-02 (*Reserve non-negativity*).

$$u \geq 0 \quad \text{and} \quad y \geq 0$$

Both virtual concentration reserve u and USDC reserve y are non-negative at all times.

CFMM-03 (*Invariant preservation on swap*).

$$\{\psi(u, y) = L \cdot C_1\} \xrightarrow{\text{swap}(\Delta u)} \{\psi(u + \Delta u, y - \Delta y) = L \cdot C_1\}$$

Every swap preserves the trading function invariant. From $\Delta y = (p_l^2/4)\Delta u$.

CFMM-04 (*Invariant preservation on mint*).

$$\{\psi(u, y) = L \cdot C_1\} \xrightarrow{\text{mint}(\Delta L)} \{\psi(u + \Delta u, y + \Delta y) = (L + \Delta L) \cdot C_1\}$$

Minting new liquidity preserves the invariant at the new liquidity level (theorem 6.10).

CFMM-05 (*Invariant preservation on burn*).

$$\{\psi(u, y) = L \cdot C_1\} \xrightarrow{\text{burn}(\Delta L)} \{\psi(u - \Delta u, y - \Delta y) = (L - \Delta L) \cdot C_1\}$$

Burning liquidity preserves the invariant at the reduced liquidity level.

CFMM-06 (*Spot price consistency*).

$$p = \frac{p_l^2}{2}\sqrt{\frac{u}{L_{\text{act}}}} = -\frac{\partial\psi/\partial u}{\partial\psi/\partial y} \cdot \frac{du}{dx}$$

The spot price derived from reserves matches the marginal rate of substitution (theorem 6.6).

CFMM-07 (*Price within tick range*).

$$p_l \leq p \leq p_u \quad \text{when } L_{\text{act}} > 0$$

The spot price stays within the active tick range $[p_l, p_u]$ when there is active liquidity.

CFMM-08 (*Liquidity additivity at tick crossing*).

$$L'_{\text{act}} = L_{\text{act}} + \Delta L_{\text{net}}$$

At each tick crossing, active liquidity updates by adding the signed net liquidity of the crossed tick. This is O(1) because ψ is 1-homogeneous in L (theorem 6.4).

CFMM-09 (*Liquidity non-negativity*).

$$L \geq 0 \quad \text{and} \quad L_{\text{act}} \geq 0$$

Total and active liquidity are non-negative.

CFMM-10 (*No-arbitrage (monotone reserves)*).

$$p' > p \;\longrightarrow\; u(p') > u(p) \;\wedge\; y(p') > y(p)$$

Both reserves are strictly increasing in price (theorem 7.1). Under Framing C, rising concentration absorbs more virtual exposure and accumulates more USDC.

CFMM-11 (*Fee accumulator monotonicity*).

$$g_{n+1}^{\text{borrow}} \geq g_n^{\text{borrow}}$$

The borrow fee growth accumulator is non-decreasing (borrow rate $\geq \gamma > 0$).

CFMM-12 (*Fee accumulator per-position*).

$$\text{fees}_i = L_i \cdot (g^{\text{borrow}} - g_{\text{baseline},i}^{\text{borrow}})$$

Per-position borrow fees use the Synthetix difference pattern [BBJ18].

CFMM-13 (*LP value bounded*).

$$0 \leq V(p) = \frac{p_u^2 - p^2}{p_l^2} \leq \frac{p_u^2 - p_l^2}{p_l^2}$$

LP value is bounded between 0 (at $p = p_u$, full loss) and the initial collateral (at $p = p_l$, no loss).

CFMM-14 (*Swap output positive*).

$$\Delta x > 0 \longrightarrow \Delta y > 0 \quad \text{and} \quad \Delta x < 0 \longrightarrow \Delta y < 0$$

Swaps in either direction produce correctly-signed outputs.

CFMM-15 (*Slippage non-negative*).

$$\frac{\Delta y}{\Delta x} = \frac{p_l^2}{4}(2x + \Delta x) \geq \frac{p_l^2}{4} \cdot 2x = p$$

The effective price is always at least the spot price (natural slippage from the $u = x^2$ transform). Equality only for infinitesimal trades.

CFMM-16 (*Coordinate transform bijection*).

$$u = x^2 \longleftrightarrow x = \sqrt{u} \quad \text{for } x \geq 0$$

The $u \leftrightarrow x$ mapping is a bijection on the non-negative reals. One `FixedPointMathLib.sqrt` call per swap.

CFMM-17 (*1-homogeneity*).

$$\psi(\lambda u, \lambda y) = \lambda \psi(u, y) \quad \forall \lambda > 0$$

The trading function is 1-homogeneous, enabling O(1) tick-based liquidity aggregation (theorem 6.4).

CFMM-18 (*Concavity*).

$$\nabla^2 \psi = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\psi(u, y) = y - (p_l^2/4)u$ is linear, hence (weakly) concave. The Hessian is zero. The trading set is convex.

CFMM-19 (*Single-sided deposit consistency*).

$$\Delta L = \frac{\Delta y \cdot p_l^2}{p_u^2 + p^2} \quad \text{and} \quad \Delta u = \Delta L \cdot \frac{4p^2}{p_l^4}$$

Single-sided USDC deposits correctly compute ΔL and virtual Δu (theorem 6.9). Only USDC enters the contract.

11.2 Replication Invariants (CFMM-20 through CFMM-25)

Source: Angeris et al. [AEC21, ACD+23].

CFMM-20 (*Replication accuracy*).

$$\{V(p)\} \xrightarrow{\text{evolve}(p \to p')} \{|\text{LP_value}(p') - V(p')| \leq \varepsilon\}$$

The CFMM LP position replicates the target payoff $V(p) = (p_u^2 - p^2)/p_l^2$ within error ε (determined by discretization and fee leakage).

CFMM-21 (*LP payoff concavity*).

$$V''(p) = -\frac{2}{p_l^2} < 0 \quad \forall\, p \in [p_l, p_u]$$

The LP payoff is strictly concave on the active domain. Required for Angeris replication theorem [AEC21].

CFMM-22 (*Reserve-payoff consistency*).

$$x(p) = -V'(p) = \frac{2p}{p_l^2} \quad \text{and} \quad y(p) = V(p) + p \cdot x(p) = \frac{p_u^2 + p^2}{p_l^2}$$

Reserves are the derivatives and Legendre transform of the payoff (theorem 6.1).

CFMM-23 (*Boundary reserves*).

$$\begin{aligned} p = p_l: &\quad x = 2/p_l,\ y = (p_u^2 + p_l^2)/p_l^2,\ V = (p_u^2 - p_l^2)/p_l^2 \\ p = p_u: &\quad x = 2p_u/p_l^2,\ y = 2p_u^2/p_l^2,\ V = 0 \end{aligned}$$

At the lower boundary, full collateral remains. At the upper boundary, collateral is fully drained (max loss for underwriter, max payout for PLP).

CFMM-24 (*Price-reserve consistency*).

$$p = \frac{p_l^2}{2} \cdot \frac{x}{L} \quad \Longleftrightarrow \quad x(p) = \frac{2pL}{p_l^2}$$

The price derived from the trading function marginal rate matches the reserve curve parameterization (theorem 6.6).

CFMM-25 (*Canonical form*).

$$\psi(u, y) = y - \frac{p_l^2}{4}\, u$$

is nondecreasing in y, nonincreasing in u, concave, and 1-homogeneous. This is the unique canonical trading function for this CFMM (Geometry paper [ACD+23], Theorem 1).

11.3 Perpetual Invariants (CFMM-26 through CFMM-28)

Source: Angeris et al. [AECB22].

CFMM-26 (*Funding rate convergence*).

$$\{|p_{\text{mark}} - p_{\text{index}}| > \delta\} \xrightarrow{\text{funding}(\Delta t)} \{|p'_{\text{mark}} - p'_{\text{index}}| < |p_{\text{mark}} - p_{\text{index}}|\}$$

The bidirectional funding mechanism drives mark-index convergence (theorem 9.8). Sensitivity $\alpha > 0$ controls speed.

CFMM-27 (*Funding rate bounds*).

$$|r_{\text{funding}n}| \leq \alpha + \lambda \cdot |\Delta p_{\text{index}}|_{\max}$$

The funding rate (including jump premium) is bounded. The bound depends on the sensitivity α, jump intensity λ, and maximum observable p_{index} change between callbacks.

CFMM-28 (*Jump-adjusted value bound*).

$$\{\Delta^+ \text{ jumps from } \delta \text{ to } \delta'\} \longrightarrow \{|\Delta\text{value}| \leq |V(\delta') - V(\delta)|\}$$

Discrete jumps in Δ^+ at liquidity events produce value changes bounded by the payoff difference (theorem 9.6).

11.4 Fee Concentration State Invariants (FCI-01 through FCI-08)

FCI-01 (*Index boundedness*).

$$0 \leq A_T \leq 1$$

The fee concentration index is bounded in $[0, 1]$ at all times. $A_T = 0$ when no positions have been removed. $A_T = 1$ is the theoretical maximum (single JIT monopoly).

FCI-02 (*Theta sum non-negativity*).

$$\Theta \geq 0$$

The aggregate turnover rate $\Theta = \sum_k \theta_k$ is non-negative since each $\theta_k = 1/\ell_k > 0$.

FCI-03 (*Position count non-negativity*).

$$N \geq 0$$

The active position count is a non-negative integer.

FCI-04 (*Null lower bound*).

$$\{N > 0\} \longrightarrow \{A_T \geq A_T^{1/N}\}$$

The real fee concentration index is always at least as large as the Ma–Crapis equal-share null (theorem 1.2). Equivalently, $\Delta^+ \geq 0$.

FCI-05 (*Deviation non-negativity*).

$$\Delta^+ = \max(0,\, A_T - A_T^{1/N}) \geq 0$$

By construction (max with zero) and by theorem 1.2.

FCI-06 (*Deviation upper bound*).

$$\Delta^+ < 1$$

Since $A_T \leq 1$ and $A_T^{1/N} \geq 0$, the deviation satisfies $\Delta^+ \leq 1$. Strict inequality holds when $N \geq 1$ (at least one position implies $A_T^{1/N} > 0$). This ensures the price mapping $p = \Delta^+/(1 - \Delta^+)$ is well-defined.

FCI-07 (*Co-primary consistency*).

$$A_T^{1/N} = \sqrt{\frac{\Theta}{N^2}} \quad \text{and} \quad \Delta^+ = \max(0,\, A_T - A_T^{1/N})$$

The derived quantities $A_T^{1/N}$ and Δ^+ are deterministic functions of the three stored state variables (A_T, Θ, N). No additional state is needed.

FCI-08 (*Liquidity event atomicity*).

$$\{\texttt{afterRemoveLiquidity}\} \longrightarrow \{A_T \text{ updated} \wedge \Theta \text{ updated} \wedge N \text{ updated}\}$$

All three co-primary state variables update atomically at each liquidity event. No intermediate state is observable where some variables are updated and others are not.

11.5 Price Mapping Invariants (FCI-09 through FCI-11)

FCI-09 (*Price non-negativity*).

$$p = \frac{\Delta^+}{1 - \Delta^+} \geq 0$$

Since $\Delta^+ \geq 0$ (FCI-05) and $1 - \Delta^+ > 0$ (FCI-06).

FCI-10 (*Price monotonicity*).

$$\Delta_1^+ < \Delta_2^+ \longrightarrow p(\Delta_1^+) < p(\Delta_2^+)$$

The price mapping is strictly increasing (theorem 1.5). Higher concentration deviation → higher insurance price.

FCI-11 (*Price-deviation invertibility*).

$$\Delta^+ = \frac{p}{1+p} \quad \text{is the unique inverse of} \quad p = \frac{\Delta^+}{1-\Delta^+}$$

The mapping is a bijection from $[0,1) \to [0,\infty)$.

11.6 FCI Perpetual Funding Invariants (FCI-12 through FCI-13)

Source: Angeris et al. [AECB22].

FCI-12 (*Funding rate convergence (FCI)*).

$$\{|p_{\text{mark}} - p_{\text{index}}| > \delta\} \longrightarrow \text{funding(liquidityEvent)} \longrightarrow \{|p'_{\text{mark}} - p'_{\text{index}}| < |p_{\text{mark}} - p_{\text{index}}|\}$$

The funding mechanism drives mark-index convergence (theorem 9.8).

FCI-13 (*Jump-adjusted value bound (FCI)*).

$$\{\Delta^+ \text{ jumps from } \delta \text{ to } \delta'\} \longrightarrow \{|\Delta\text{value}| \leq |V(\delta') - V(\delta)|\}$$

Discrete jumps in Δ^+ at liquidity events do not cause value changes exceeding the payoff difference (theorem 9.6).

11.7 Insurance CFMM Invariants (INS-01 through INS-10)

INS-01 (*Collateral solvency*).

$$y \geq L \cdot \frac{p_u^2 + p^2}{p_l^2} \quad \text{(USDC reserve covers all LP positions)}$$

The USDC reserve is sufficient to cover the numeraire reserve requirement at the current price. Underwriters cannot lose more than their deposit (theorem 7.3).

INS-02 (*Utilization bound*).

$$0 \leq U_a = \frac{L_{\text{borrowed}}}{L_{\text{total}}} \leq 1$$

Utilization is bounded in $[0, 1]$. $L_{\text{borrowed}} \leq L_{\text{total}}$ is enforced by reverting borrows that would exceed available liquidity.

INS-03 (*Borrow rate positivity*).

$$r_{\text{borrow}}(U_a) \geq \gamma > 0 \quad \forall\, U_a \in [0, 1]$$

The borrow rate is always at least γ (the base premium floor). Underwriters always earn a positive rate (theorem 8.2).

INS-04 (*Borrow rate monotonicity*).

$$U_{a1} < U_{a2} \;\longrightarrow\; r_{\text{borrow}}(U_{a1}) \leq r_{\text{borrow}}(U_{a2})$$

The borrow rate is non-decreasing in utilization. Higher utilization means higher cost, incentivizing new underwriter deposits or PLP exit (theorem 8.2).

INS-05 (*Separate accumulator accounting*).

$$g^{\text{borrow}} \geq 0 \quad \text{(always positive)} \qquad g^{\text{funding}} \in \mathbb{R} \quad \text{(bidirectional)}$$

The borrow and funding accumulators are maintained separately (theorems 8.3 and 9.7). The borrow accumulator is non-negative (PLP always pays underwriter). The funding accumulator can be positive or negative (bidirectional). Conflating them breaks funding convergence [AECB22].

INS-06 (*Max-price tracking*).

$$p_{\text{max},i} = \max_{t \in [\text{mint}_i, \text{now}]} p_{\text{index}}(t) \quad \text{and} \quad p'_{\text{max},i} \geq p_{\text{max},i}$$

Each PLP position stores the running maximum of p_{index} since mint. Updated at every hook callback when $p_{\text{index}} > p_{\text{max},i}$. This value is monotone non-decreasing.

INS-07 (*Payout formula*).

$$\text{payout}_i = \text{premium}_i \times \left[\left(\frac{p_{\text{max},i}}{p_l} \right)^2 - 1 \right]^+$$

The PLP payout at exit follows the power-2 payoff. The plus operator $[\cdot]^+$ ensures non-negative payouts. The premium is defined as $\text{premium}_i = \gamma \cdot \text{lifetime_fees}_i + \text{accrued_borrow}_i + \text{accrued_funding}_i$.

INS-08 (*Payout cap*).

$$\text{payout}_i \leq \text{premium}_i \times \left[\left(\frac{p_u}{p_l} \right)^2 - 1 \right]$$

The payout is capped by the tick range. Since $p_{\max,i} \leq p_u$ within the CLAMM range, the payout cannot exceed the cap. No global cap parameter is needed—the per-position tick range enforces bounded loss for underwriters.

INS-09 (*Initialization invariant*).

$$\psi(u_0, y_0) = L_0 \cdot \frac{p_u^2}{p_l^2}$$

At initialization (first deposit), the invariant holds with $C_1 = p_u^2/p_l^2$ (theorem 10.3).

INS-10 (*Exit settlement conservation*).

$$\{\texttt{exit}_i\} \longrightarrow \{L'_{\text{borrowed}} = L_{\text{borrowed}} - L_i\} \wedge \{\text{net}_i = \max(\text{payout}_i - \text{premium}_i, 0)\}$$

At PLP exit: borrowed liquidity returns to the pool (utilization decreases), and the PLP receives the net settlement (payout minus premium, floored at zero). If payout < premium, the excess stays in reserves as underwriter profit.

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